**SAVEETHA SCHOOL OF ENGINEERING**

**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES**

**COMPUTER SCIENCE AND ENGINEERING**

| | | |
|---|---|---|
| **BRANCH** | : | CSE |
| **YEAR/SEM** | : | II & III/ IV & VI |
| **SUB CODE & TITLE** | : | **CSA1322 – Theory of Computation** |
| **SUBMISSION DATE** | : | **15.07.2026** |

## Activity 1: DFA Design

**Activity 1: DFA Design – Even Number of 1s**

**Problem Statement:**

Design a Deterministic Finite Automaton (DFA) that accepts all binary strings over Σ = {0,1} which contain an even number of 1s.

**Student Tasks:**

1. Define the DFA as
   M = (Q, Σ, δ, $q_0$, F)
2. Draw the state transition diagram.
3. Identify accepting and rejecting states.
4. Test the DFA for inputs:
   - 1010
   - 111
   - 1100

**Expected Learning:**

Understanding how states represent parity (even/odd) and how DFA tracks memory using states.

SOLUTION:

**1. Define the DFA**
The DFA is defined as

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where

- **Q = {$q_0$, $q_1$}**

- **Σ = {0,1}**
- **$q_0$ = Start State**
- **F = {$q_0$}** (Accepting State)

**Transition Function (δ)**

| Present State | Input 0 | Input 1 |
|---|---|---|
| →$q_0$ | $q_0$ | $q_1$ |
| $q_1$ | $q_1$ | $q_0$ |

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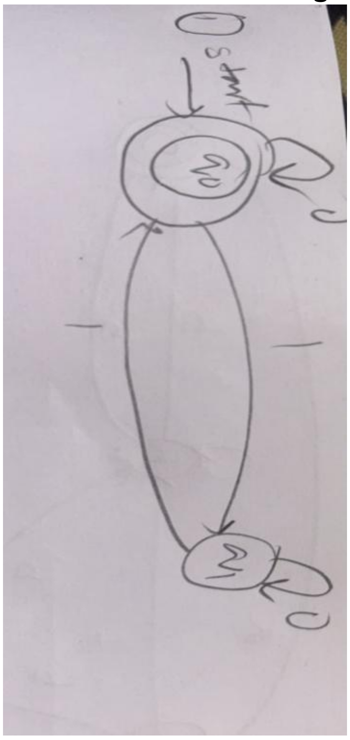
## 2. State Transition Diagram



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## 3. Accepting and Rejecting States

**Accepting State**

- **$q_0$** → Even number of **1s**

**Rejecting State**

- **$q_1$** → Odd number of **1s**

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## 4. Testing the DFA

**Input 1: 1010**

| Input | Current State |
|---|---|
| Start | $q_0$ |
| 1 | $q_1$ |
| 0 | $q_1$ |
| 1 | $q_0$ |
| 0 | $q_0$ |

Final State = **$q_0$**

✅ **Accepted**

(Number of 1s = 2 → Even)

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**Input 2: 111**

| Input | Current State |
|---|---|
| Start | $q_0$ |
| 1 | $q_1$ |

| Input | Current State |
| --- | --- |
| 1 | $q_0$ |
| 1 | $q_1$ |

Final State = $\mathbf{q_1}$

❌ **Rejected**

(Number of 1s = 3 → Odd)

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**Input 3: 1100**

| Input | Current State |
| --- | --- |
| Start | $q_0$ |
| 1 | $q_1$ |
| 1 | $q_0$ |
| 0 | $q_0$ |
| 0 | $q_0$ |

Final State = $\mathbf{q_0}$

✅ **Accepted**

(Number of 1s = 2 → Even)

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**Result**

| Input String | Number of 1s | Final State | Result |
| --- | --- | --- | --- |
| 1010 | 2 | $q_0$ | ✅ Accepted |
| 111 | 3 | $q_1$ | ❌ Rejected |
| 1100 | 2 | $q_0$ | ✅ Accepted |

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**Expected Learning**

- The DFA uses **two states** to remember whether the number of **1s** seen so far is **even** or **odd**.
- Reading a **0** does **not** change the parity, so the DFA stays in the same state.
- Reading a **1** changes the parity, so the DFA switches between $\mathbf{q_0}$ **(Even)** and $\mathbf{q_1}$ **(Odd)**.
- A string is **accepted** only if it ends in $\mathbf{q_0}$, meaning it contains an **even number of 1s**.

**Activity 2: DFA for Strings Ending with “01”**

**Problem Statement:**

Construct a DFA over Σ = {0,1} that accepts all strings that end with “01”.

**Student Tasks:**

1. Design the DFA with minimum number of states.
2. Draw the transition diagram.
3. Write the transition table.
4. Trace the DFA for inputs:
   - 01
   - 101
   - 1101
   - 100

**Expected Learning:**
Understanding pattern matching using DFA and suffix tracking.

SOLUTION:

**1. Design the DFA (Minimum Number of States)**

The DFA is defined as

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where

- **Q = {$q_0$, $q_1$, $q_2$}**
- **Σ = {0,1}**
- **$q_0$ = Start State**
- **F = {$q_2$}** (Accepting State)

**Meaning of States**

- **$q_0$** → No useful suffix matched.
- **$q_1$** → Last symbol is **0**.
- **$q_2$** → Last two symbols are **01** (Accept State).

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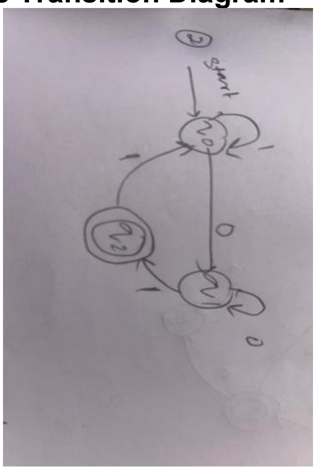
**2. State Transition Diagram**

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**3. Transition Table**

| Present State | Input 0 | Input 1 |
|---|---|---|
| →$q_0$ | $q_1$ | $q_0$ |

| Present State | Input 0 | Input 1 |
|---|---|---|
| $q_1$ | $q_1$ | $q_2$ |
| *$q_2$ | $q_1$ | $q_0$ |

---

**4. Trace the DFA**

**Input 1: 01**

| Input | State |
|---|---|
| Start | $q_0$ |
| 0 | $q_1$ |
| 1 | $q_2$ |

Final State = $\mathbf{q_2}$

✅ **Accepted**

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**Input 2: 101**

| Input | State |
|---|---|
| Start | $q_0$ |
| 1 | $q_0$ |
| 0 | $q_1$ |
| 1 | $q_2$ |

Final State = $\mathbf{q_2}$

✅ **Accepted**

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**Input 3: 1101**

| Input | State |
|---|---|
| Start | $q_0$ |
| 1 | $q_0$ |
| 1 | $q_0$ |
| 0 | $q_1$ |
| 1 | $q_2$ |

Final State = $\mathbf{q_2}$

✅ **Accepted**

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**Input 4: 100**

| Input | State |
|---|---|
| Start | $q_0$ |
| 1 | $q_0$ |
| 0 | $q_1$ |
| 0 | $q_1$ |

Final State = $\mathbf{q_1}$

❌ **Rejected**

(The string ends with **00**, not **01**.)

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**Result Table**

| Input String | Ends With | Final State | Result |
|---|---|---|---|
| 01 | 01 | $q_2$ | ✅ Accepted |
| 101 | 01 | $q_2$ | ✅ Accepted |
| 1101 | 01 | $q_2$ | ✅ Accepted |
| 100 | 00 | $q_1$ | ❌ Rejected |

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**Expected Learning**

- The DFA **tracks the last one or two symbols** of the input.
- It reaches the accepting state $\mathbf{q_2}$ only when the input **ends with "01"**.
- Any additional input changes the state according to the transition function, ensuring **only strings whose final two symbols are "01" are accepted**.

**Activity 3: Identify Accept/Reject from Given DFA**

**Problem Statement:**

Given a DFA M that accepts strings over $\Sigma = \{a, b\}$ which contain at least one 'a'.

**Student Tasks:**

1. Identify the final (accepting) state.
2. For the following inputs, say Accept or Reject and show the transition path:
   - bbbb
   - abbb
   - bab
   - ε (empty string)
3. Write the language accepted by the DFA in words.

**Expected Learning:**

Tracing input strings through a DFA and understanding acceptance condition.

SOLUTION:

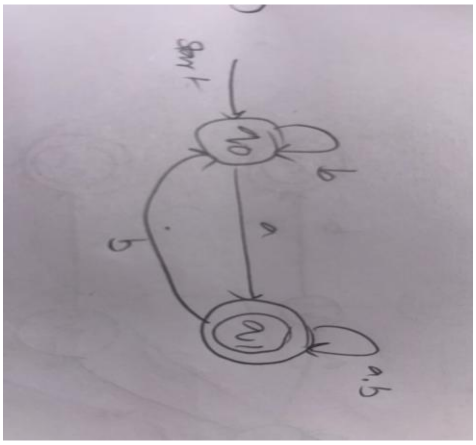
**1. Identify the Final (Accepting) State**

**DFA Definition**

States:

- $\mathbf{q_0}$ = No **'a'** has been seen yet (**Start State**)
- $\mathbf{q_1}$ = At least one **'a'** has been seen (**Final State**)

**DFA Diagram**




- **Start State:** → $q_0$
- **Accepting State:** ⭐ $q_1$

## 2. Test the Given Inputs

### (a) Input = bbbb

Transition Path:

q0 --b--> q0

q0 --b--> q0

q0 --b--> q0

q0 --b--> q0

Final State = $\mathbf{q_0}$

❌ **Rejected**

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### (b) Input = abbb

Transition Path:

q0 --a--> q1

q1 --b--> q1

q1 --b--> q1

q1 --b--> q1

Final State = $\mathbf{q_1}$

✅ **Accepted**

---

### (c) Input = bab

Transition Path:

q0 --b--> q0

q0 --a--> q1

q1 --b--> q1

Final State = $\mathbf{q_1}$

✅ **Accepted**

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### (d) Input = ε (Empty String)

Transition Path:

q0

Final State = $\mathbf{q_0}$

❌ **Rejected**

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## 3. Language Accepted by the DFA

### In Words

**The DFA accepts all strings over the alphabet {a, b} that contain at least one 'a'.**

### Mathematical Representation

$$L = \{\, w \in \{a, b\}^* \mid w \text{ contains at least one } a \,\}$$

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**Result Table**

| Input | Final State | Result |
| --- | --- | --- |
| bbbb | $q_0$ | ❌ Rejected |
| abbb | $q_1$ | ✅ Accepted |
| bab | $q_1$ | ✅ Accepted |
| ε | $q_0$ | ❌ Rejected |

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**Expected Learning**

- Learn how to trace strings through a DFA.
- Understand why a string is accepted only after reading **at least one 'a'**.
- Identify accepting and rejecting states.

**Activity 4: Real-Time Application DFA – Password Rule**

**Problem Statement**:
Design a DFA that accepts all strings over Σ = {a, b} that
✓ Start with ‘a’ and
✓ End with ‘b’

**Student Tasks:**

1. Draw the DFA.
2. Clearly mark start and final states.
3. Check whether the following are accepted or rejected:
   - ab
   - aab
   - abb
   - ba
   - a

**Expected Learning:**
Applying DFA concepts to real-world constraints like validation rules.

SOLUTION:

**1. Define the DFA**

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where:

- **Q = {$q_0$, $q_1$, $q_2$, qd}**
- **Σ = {a,b}**
- **$q_0$ = Start State**
- **F = {$q_2$}**

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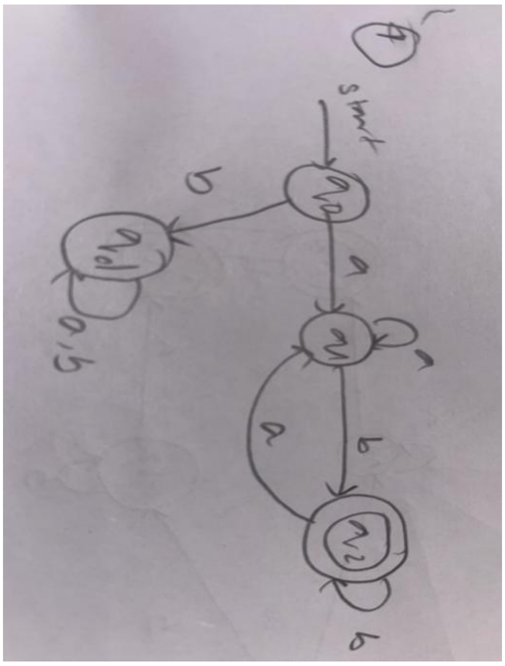
**2. DFA Diagram**



q0 --b--> qd

qd --a,b--> qd

---

### 3. Transition Table

| Present State | a | b |
|---|---|---|
| $\rightarrow q_0$ | $q_1$ | qd |
| $q_1$ | $q_1$ | $q_2$ |
| $*q_2$ | $q_1$ | $q_2$ |
| qd | qd | qd |

---

### 4. Start and Final States

- **Start State:** $\rightarrow q_0$
- **Final State:** ⭐ $q_2$

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### 5. Check the Given Inputs

**(a) Input = ab**

q0 → q1 → q2

✅ **Accepted**

---

**(b) Input = aab**

q0 → q1 → q1 → q2

✅ **Accepted**

---

**(c) Input = abb**

q0 → q1 → q2 → q2

✅ **Accepted**

---

**(d) Input = ba**

q0 → qd → qd

❌ **Rejected**

---

**(e) Input = a**

q0 → q1

❌ **Rejected**

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**Result Table**

| Input | Starts with 'a' | Ends with 'b' | Result |
|---|---|---|---|
| ab | Yes | Yes | ✅ Accepted |
| aab | Yes | Yes | ✅ Accepted |

| Input | Starts with 'a' | Ends with 'b' | Result |
|---|---|---|---|
| abb | Yes | Yes | ✅ Accepted |
| ba | No | No | ❌ Rejected |
| a | Yes | No | ❌ Rejected |

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**Expected Learning**

- Learn to design a DFA based on multiple conditions.
- Understand how DFAs can be used in **real-world validation**, such as checking password rules.
- Practice tracing strings through the DFA to determine acceptance or rejection.